Who Reaches the Ph.D.? SC, ST, OBC and Women in Indian Higher Education

Inclusion Study, IISER Kolkata iGEM team

Background

Access to higher education in India has widened, but the share of students from SC, ST and OBC communities falls at the Ph.D. stage.

Objectives

To measure, in a quantitative manner, how the share of SC, ST and OBC students and of women changes from undergraduate to Ph.D. level in India, and how it varies by state and type of institution.

Results

- A dataset was built from four years of the All India Survey on Higher Education (AISHE, 2019-20 to 2023-24) covering enrolment by level, social category, gender, state and university type
- We find that the SC share falls from 15.6% at undergraduate level to 10.0% at Ph.D. level, and the ST share from 6.3% to 4.1% (2023-24)
- We also found that women reach parity at Ph.D. level nationally (50.1%) but are only 44.4% of SC Ph.D. students, and that the gap between women's undergraduate and Ph.D. share is more than 11 points in some states
- Measured the step-down as a retention index (0.64 for SC and 0.65 for ST in 2023-24, against 0.68 and 0.63 in 2019-20) and as people: about 18,700 SC and 7,400 ST Ph.D. students short of their UG shares
- Found that SC women are the thinnest group (44.4% of SC Ph.D. students), that Ph.D. students per 100 PG students range from 2.3 to 10.4 between large states, and that women's share in INI Ph.D. programmes lags all universities by 14 points, a gap that grew over five years
- Identified what public data cannot show, and made recommendations for publishing Ph.D. data by category

Key Findings (brief overview):

1. **The step-down happens at the Ph.D.** In 2023-24 the SC share is 15.6% (UG), 13.7% (PG), 10.0% (Ph.D.); ST 6.3%, 6.2%, 4.1%; OBC 41.0%, 37.5%, 28.1%.

2. **It has not narrowed.** The retention index (Ph.D. share divided by UG share) was 0.68 for SC in 2019-20 and 0.64 in 2023-24; for ST 0.63 and 0.65; for OBC 0.66 and 0.69. Women went from 0.92 to 1.01.
3. **In people:** had Ph.D. shares matched UG shares, there would be about 18,700 more SC, 7,400 more ST and 43,100 more OBC Ph.D. students in 2023-24 (an illustration of scale, not a target).
4. **SC women are the thinnest group.** Women are 50.1% of all Ph.D. students but 44.4% of SC and 47.6% of ST Ph.D. students. The SC-women gap was 5.9 points in 2019-20 and 5.7 in 2023-24.
5. **States differ widely.** SC share of enrollment runs from 6.9% (Kerala) to 25.4% (Punjab) among large states; ST from 0.6% to 20.6%. Ph.D. students per 100 PG students run from 2.3 (Madhya Pradesh) to 10.4 (Assam, Punjab); India is 5.9.
6. **Women's Ph.D. share is rising almost everywhere** (all 12 large disciplines since 2021-22), but Engineering and Technology is still 37% women, and in Institutes of National Importance women are 34.6% of Ph.D. students, 14.0 points below all universities (the gap was 11.8 in 2019-20).
7. **Private universities have the lowest SC and ST shares** (state private: 7.4% SC, 4.4% ST; private deemed: 5.6% SC, 1.3% ST). Central universities are below the 15%, 7.5% and 27% reference points on all three (13.2%, 4.4%, 18.0%).
8. **Out-turn trails enrolment.** In 2023-24 the SC share of students passing out was 92% of the SC share of enrolment (ST 94%, OBC 97%).

All numbers are shares of recorded enrolment. Counts jump in 2023-24 because of wider survey coverage, so years are compared by share.

Introduction

Higher education in India is often described with one enrolment number. That number hides where students are lost. Reservation in admissions to central institutions sets reference points of 15% for Scheduled Castes (SC), 7.5% for Scheduled Tribes (ST) and 27% for Other Backward Classes (OBC), but the share of students from these groups does not stay constant as people move from undergraduate (UG) to postgraduate (PG) to Ph.D. level.

The Ph.D. is the stage from which people become supervisors, reviewers and team leaders in science. If some communities are thinner at that stage, they are thinner in the rooms where research questions are chosen.

Gender follows a different pattern. Women are about half of UG students and more than half of PG students in India, so a single national figure suggests the problem is solved. A closer look by state, by discipline and by institution type shows it is not.

We therefore followed a data-driven approach, in the spirit of earlier iGEM gender studies: use public data, show every number, and let anyone test their own hypothesis. Because our own work is in basic science, we paid attention to the Ph.D. stage and to science and engineering disciplines.

Data and Method: a public database, and its limits

Online data

All numbers come from the Final Reports of the All India Survey on Higher Education (AISHE), published by the Department of Higher Education, Ministry of Education, at aishe.gov.in. We used the Excel versions for 2019-20, 2021-22, 2022-23 and 2023-24. The documents are attached in the wiki for reference. The tables used report enrolment by programme, level, social category (SC, ST, OBC) and gender, by state, and by type of university.

Level grouping

AISHE reports category data by programme, not by level. We grouped programmes by name into undergraduate (for example B.A., B.Sc., B.Tech.), postgraduate (M.A., M.Sc., M.Tech.) and Ph.D. (the "Ph.D.-Doctor of Philosophy" programme). This covers about 91% of all enrolment in each year. The full mapping is in the downloadable database so that anyone can check it.

Database

For each year we recorded: enrolment by level, total, women, SC, ST and OBC; the same by state; and by type of university, including Institutes of National Importance (IITs, NITs, IISERs, IIMs, AIIMS and others, reported together). Shares are calculated by formula from the counts. Totals were cross-checked: the sum of states equals the all-India figure in every state table.

Companion workbook: AISHE_inclusion_analysis.xlsx (attached)

What we compare

The main measure is the share of each group among all students at a level, compared across levels in the same year. A retention index divides the Ph.D. share by the UG share: 1.00 means no drop. This compares a stock of Ph.D. students admitted over several years with the current UG stock, so it is an indicator only, not a study that follows a cohort.

The Ph.D. pipeline: SC, ST and OBC shares fall at every step

In 2023-24, SC students were 15.6% of UG enrolment, 13.7% of PG enrolment and 10.0% of Ph.D. enrolment. ST students were 6.3%, 6.2% and 4.1%. OBC students were 41.0%, 37.5% and 28.1%. Women do not follow this pattern: 49.5%, 55.7% and 50.1%.

The retention index (Ph.D. share divided by UG share) is 0.64 for SC, 0.65 for ST and 0.69 for OBC, against 1.01 for women. At Ph.D. level the SC share is 5.0 percentage points below the 15% reference point and the ST share is 3.4 points below 7.5%.

For both groups the larger step down is from PG to Ph.D.: SC from 13.7% to 10.0% (against 15.6% to 13.7% from UG to PG) and ST from 6.2% to 4.1% (against 6.3% to 6.2%).

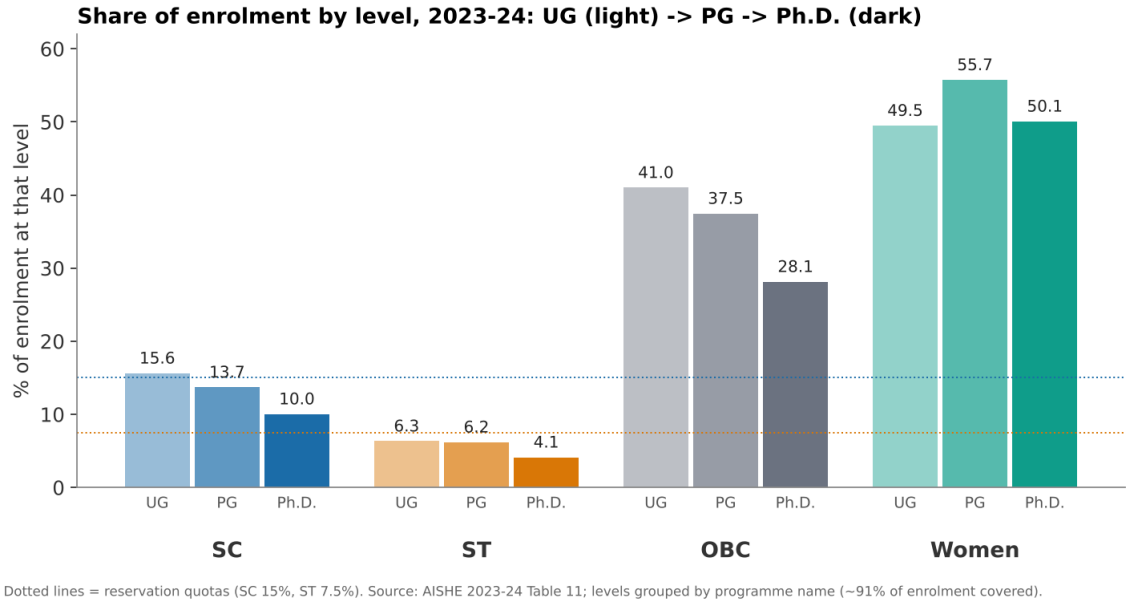


Figure 1: *Share of enrollment by level, 2023-24. Each group is shown at UG (light), PG and Ph.D. (dark) level. Dotted lines mark the 15% SC and 7.5% ST reference points. Source: AISHE 2023-24, Table 11.*

These shares have not moved since 2019

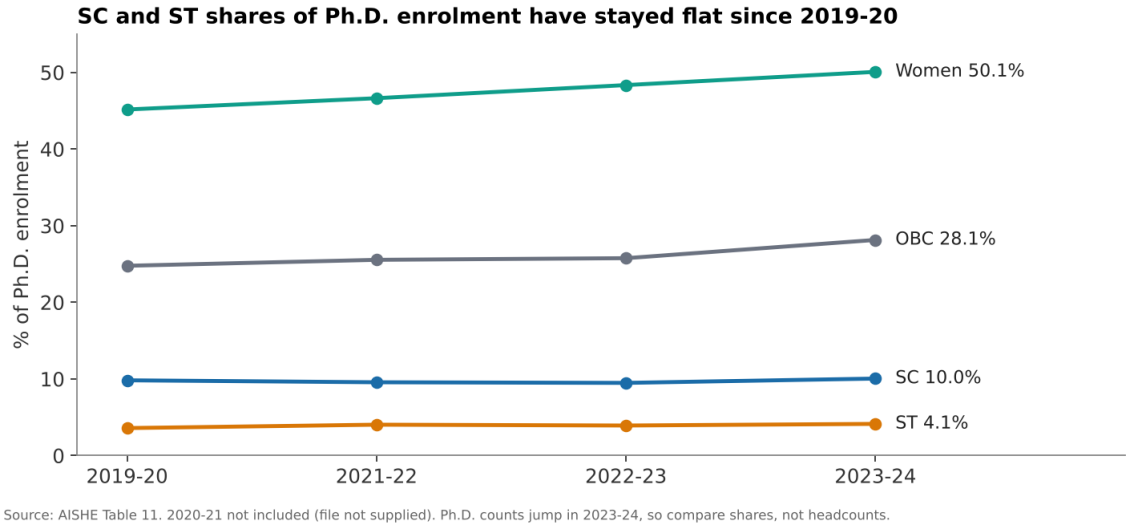


Figure 2: *Ph.D. enrolment shares, 2019-20 to 2023-24. The SC share was 9.8% in 2019-20 and 10.0% in 2023-24; ST was 3.6% and 4.1%. Women rose from 45.2% to 50.1%. The 2020-21 year is not shown because the full programme-wise table was not part of our files. Source: AISHE Table 11.*

Table 1: *Ph.D. students by social category, national*

Year	All	Women	SC	ST	OBC
2023-24	334,764	167,638	33,637	13,843	94,210
2022-23	227,131	109,818	21,546	8,912	58,506
2021-22	208,433	97,208	19,942	8,400	53,246
2019-20	199,225	89,976	19,571	7,149	49,346

Counts jump 47% between 2022-23 and 2023-24. This is an extremely huge jump and we're not sure of the reasons behind the same. We tried to find the reasoning behind the same but were unable to find a satisfactory answer. Hence, we tentatively suggest that this is probably the result of a wider survey coverage and not real growth. Source: AISHE Table 11.

Among SC Ph.D. students 44.4% are women (14,922 of 33,637); among ST, 47.6% (6,595 of 13,843). For all Ph.D. students women are 50.1%, so SC women are the most under-represented group in this table.

Is the drop getting smaller? The retention index

The retention index divides a group's share of Ph.D. enrolment by its share of UG enrolment in the same year; 1.00 means no drop. Over four survey years the SC index has stayed between 0.61 and 0.68 and the ST index between 0.62 and 0.65. OBC moved from 0.66 to 0.69. Women moved from 0.92 to 1.01.

The SC Ph.D. share went from 9.8% to 10.0%, which looks like stagnation. Part of the reason is that the SC share of UG enrolment itself rose (14.6% to 15.6%); Ph.D. shares reflect admissions over several years and respond slowly.

The index can be split into two steps. From UG to PG it is 0.88 for SC, 0.98 for ST and 0.91 for OBC. From PG to Ph.D. it is 0.73, 0.67 and 0.75. For ST the loss is almost entirely at the PG-to-Ph.D. step.

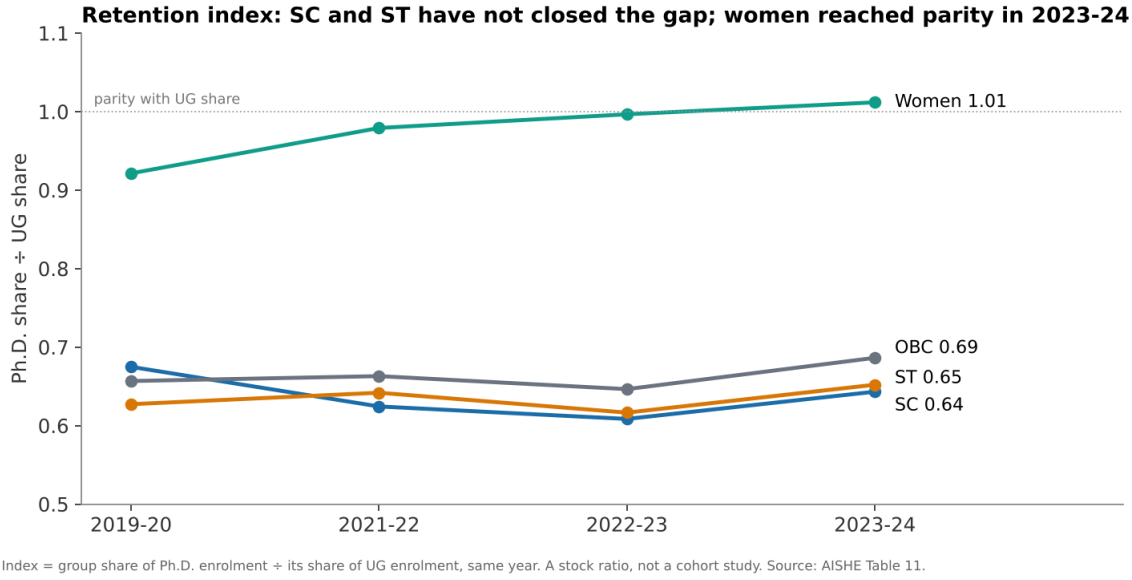


Figure 3: *Retention index, 2019-20 to 2023-24. Index = share of Ph.D. enrolment divided by share of UG enrolment, same year. Source: AISHE Table 11.*

The step-down in people

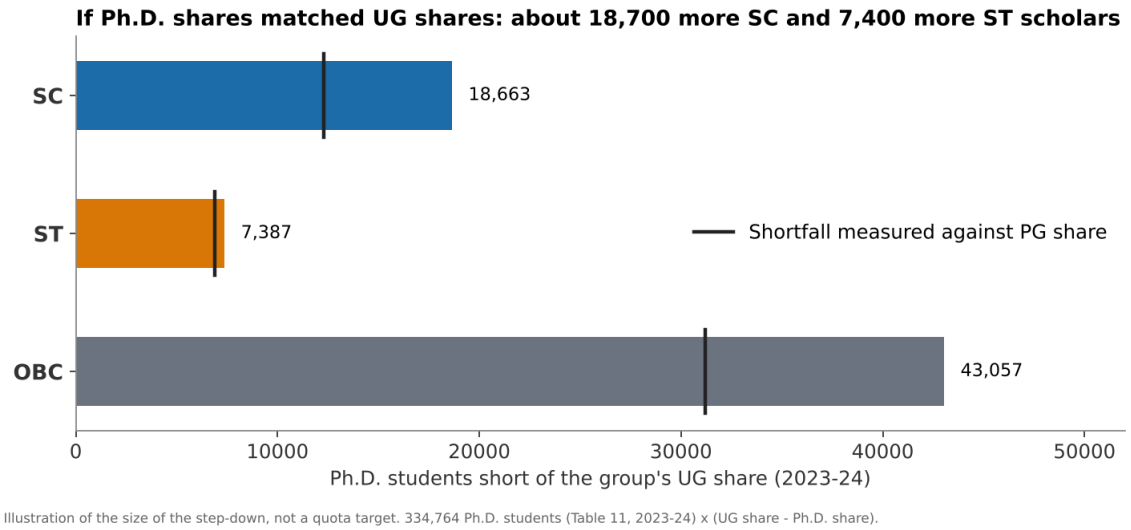


Figure 4: *How many Ph.D. students would the UG share imply? Bars: shortfall of actual Ph.D. enrolment against the group's UG share. Black ticks: shortfall against the PG share. Source: AISHE Table 11, our calculation.*

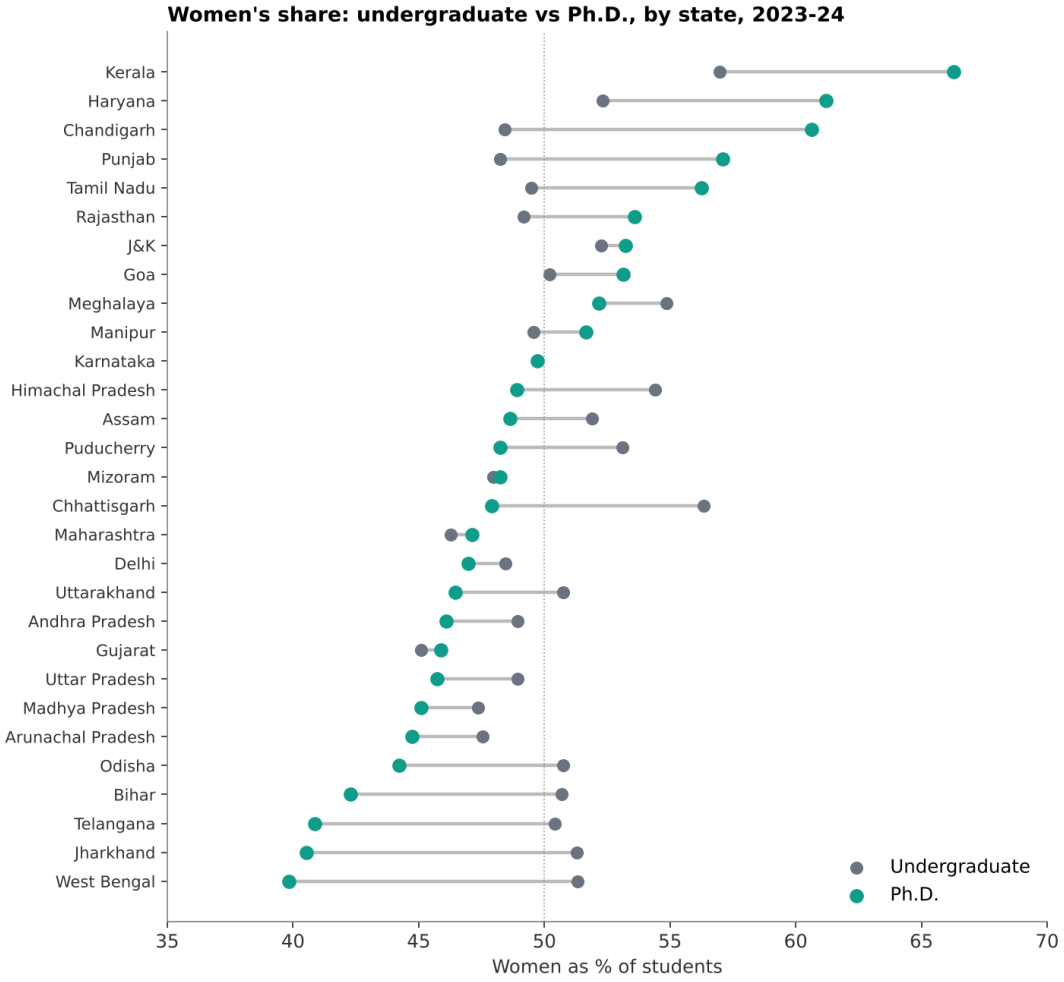
In 2023-24 there were 334,764 Ph.D. students in Table 11. At the UG share, SC students would number about 52,300 instead of 33,637 (a shortfall of 18,700). For ST the figures are 21,200 against 13,843 (7,400) and for OBC 137,300 against 94,210 (43,100). Against the PG share the shortfalls are smaller: 12,300 SC, 6,900 ST and 31,200 OBC.

This is an illustration of the size of the step-down, not a target or an estimate of discrimination. The Ph.D. stock includes students admitted in earlier years when UG shares were lower, and some Ph.D. places fall outside reservation rules.

Women: parity on average, large gaps by state and by field

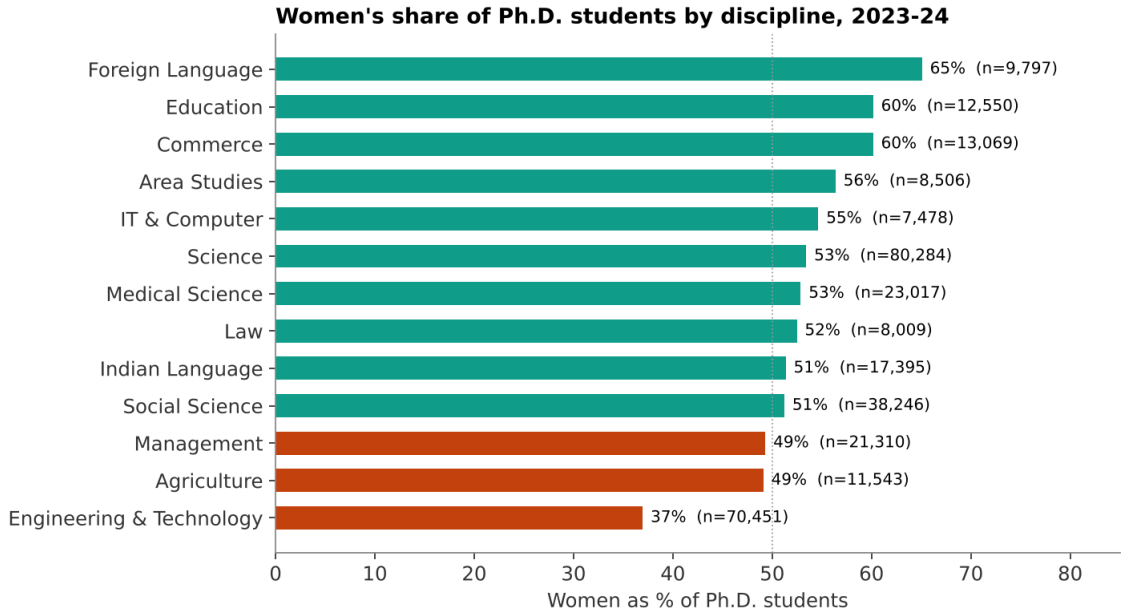
Women are 49.5% of UG students and 50.1% of Ph.D. students nationally. The average hides large differences between states. In West Bengal women are 51.3% of UG students but 39.8% of Ph.D. students. The same drop appears in Jharkhand (51.3% to 40.5%), Telangana (50.4% to 40.9%), Bihar (50.7% to 42.3%) and Odisha (50.7% to 44.2%). In Kerala (66.3%), Haryana (61.2%), Chandigarh (60.6%), Punjab (57.1%) and Tamil Nadu (56.3%), women have a larger share at Ph.D. than at UG level.

By discipline, Engineering and Technology has the lowest share of women among Ph.D. students (37% of 70,451), while Science is at 53% (80,284 students). Foreign languages, education and commerce are 60% to 65% women.



Source: AISHE 2023-24 Table 43/6 (estimated). States with fewer than 1,000 Ph.D. students omitted.

Figure 5: *Women's share, undergraduate vs Ph.D., by state, 2023-24. States with fewer than 1,000 Ph.D. students are omitted. Source: AISHE Table 43 (estimated).*

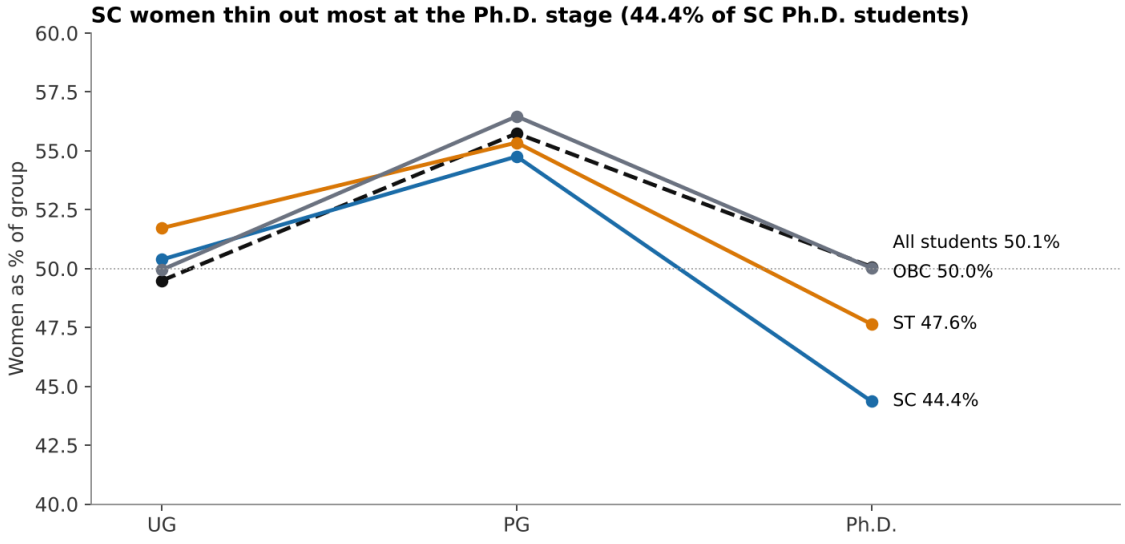


Source: AISHE 2023-24 Table 13. Disciplines with at least 3,000 Ph.D. students. Orange = below parity.

Figure 6: *Women's share of Ph.D. students by discipline, 2023-24. Disciplines with at least 3,000 Ph.D. students. Orange bars are below parity. Source: AISHE Table 13.*

In Institutes of National Importance taken together, women are far fewer at every level: 26.5% at UG, 30.9% at PG and 34.6% at Ph.D. in 2023-24. The Ph.D. share has risen from 32.2% in 2019-20. AISHE does not separate IITs from IISERs, so this cannot yet be said for IISERs alone.

SC and ST women



Women as % of each group at that level, 2023-24. Source: AISHE Table 11 (actual responses).

Figure 7: *Women as a share of each group, by level, 2023-24. At UG level women are about half of every group. At the Ph.D. stage SC women drop to 44.4%. Source: AISHE Table 11.*

At UG level women are 49.5% of all students, 50.4% of SC, 51.7% of ST and 50.0% of OBC students. At PG level they are a majority in every group (54.8% to 56.5%). At Ph.D. level the groups separate: OBC women are 50.0%, ST women 47.6% and SC women 44.4%.

SC women fall 10.4 points from PG to Ph.D., against 5.6 points for all students. The gap between SC women and all women at Ph.D. level has barely moved: 5.9 points in 2019-20 and 5.7 in 2023-24. In headcount there are 14,922 SC and 6,595 ST women among Ph.D. students in 2023-24.

Women by discipline over time

Between 2021-22 and 2023-24 women's share of Ph.D. students rose in all 12 disciplines with at least 8,000 Ph.D. students. The biggest gains were in Foreign Language (60% to 65%), Area Studies (52% to 56%) and Social Science (47% to 51%). Agriculture and Management are close to parity (49%).

Engineering and Technology remains far below the rest, at 37% of 70,451 Ph.D. students, up from 34% two years earlier. It is the second-largest discipline by Ph.D. enrollment after Science (80,284), so it pulls the national figure down.

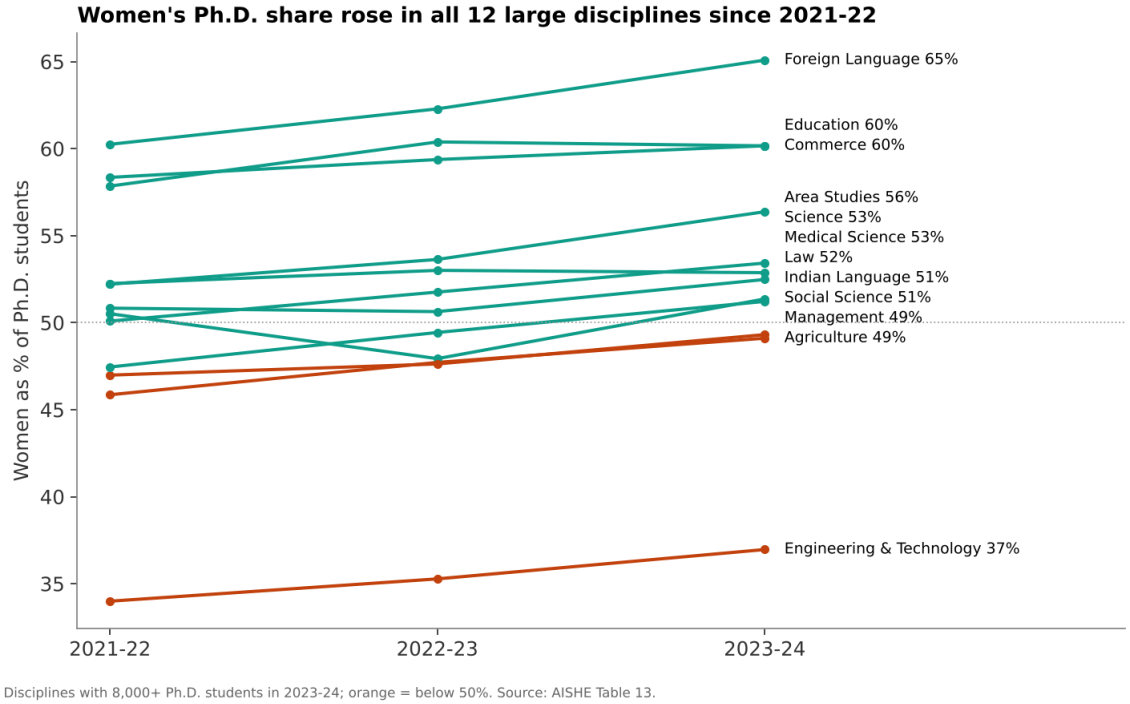


Figure 8: *Women's share of Ph.D. students, large disciplines, 2021-22 to 2023-24. Disciplines with 8,000 or more Ph.D. students in 2023-24; orange lines end below 50%. Source: AISHE Table 13.*

States: where SC and ST students are already thin

AISHE does not publish SC or ST by state at the Ph.D. level, so state figures cover all levels together. They show where SC and ST students are thin before the Ph.D. stage. Among the 20 states with at least 5 lakh students, the SC share of enrolment runs from 6.9% in Kerala to 25.4% in Punjab. All India is 15.5% SC, 6.4% ST and 40.1% OBC.

ST share is highest in Jharkhand (20.6%), Chhattisgarh (20.1%) and Assam (18.1%). Between 2019-20 and 2023-24 the SC share rose in 18 of the 20 larger states, most in Bihar (+2.9 points), Haryana (+2.8) and Punjab (+2.6), and fell in Delhi (-0.7) and Madhya Pradesh (-0.3).

**Note: These shares are not adjusted for each state's SC and ST population.*

Table 2: *SC, ST and OBC share of enrolment, all levels, 2023-24*

State	Students (lakh)	SC	ST	OBC	Change in SC share since 2019-20 (points)
Kerala	12.8	6.9%	1.1%	54.8%	+0.5
Assam	7.9	9.3%	18.1%	26.6%	+1.0
Jharkhand	9.2	9.4%	20.6%	43.0%	+0.5
Gujarat	19.4	9.5%	12.0%	32.9%	+0.4
Maharashtra	46.5	12.4%	4.8%	31.3%	+0.6
Delhi	12.3	13.6%	1.7%	16.0%	-0.7
Rajasthan	27.4	17.6%	12.5%	43.1%	+1.6
Odisha	11.9	17.8%	15.4%	25.8%	+1.7
Tamil Nadu	35.8	18.6%	0.9%	67.7%	+1.2
West Bengal	24.5	18.9%	3.8%	18.5%	+0.6
Uttar Pradesh	72.7	19.0%	1.0%	43.9%	+0.6
Punjab	9.3	25.4%	0.6%	13.5%	+2.6

The six lowest and six highest SC shares among the 20 states with at least 5 lakh students. Source: AISHE Table 45 (estimated).

Where SC and ST shares moved, 2019-20 to 2023-24

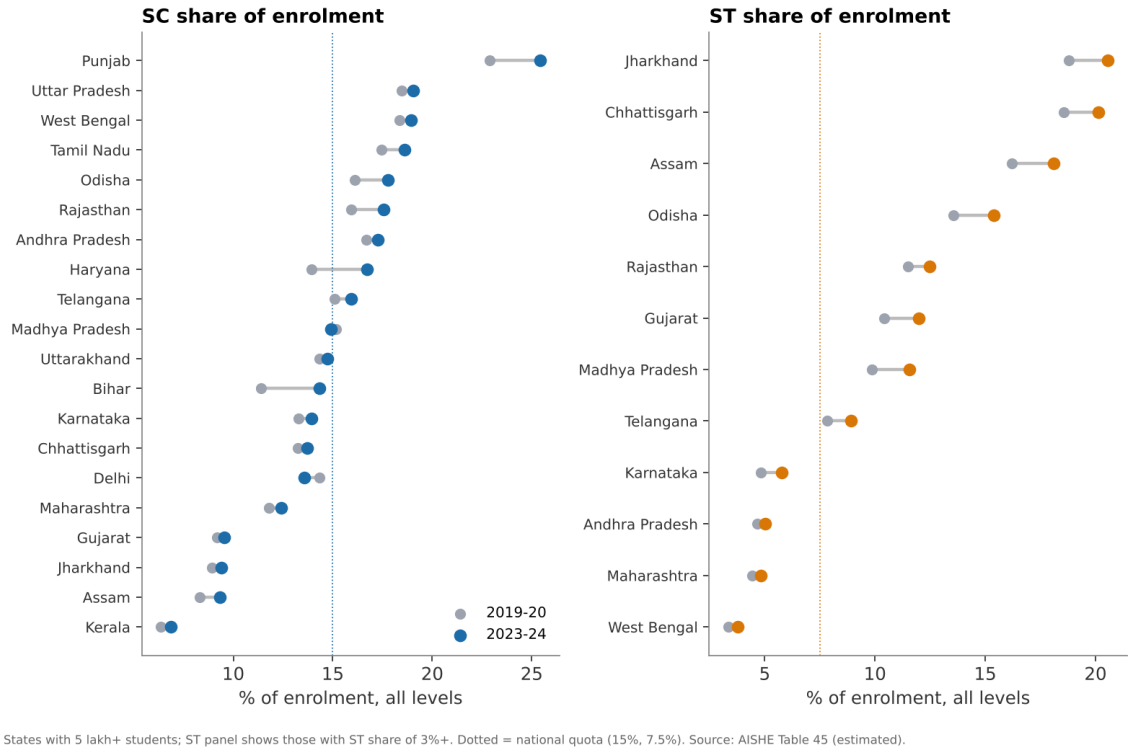


Figure 9: *SC and ST share of enrolment by state, 2019-20 and 2023-24. States with at least 5 lakh students; the ST panel shows states with 3% or more ST students. Dotted lines mark 15% and 7.5%. Source: AISHE Table 45 (estimated).*

SC share is above 15% in 9 of the 20 larger states, led by Punjab (25.4%), Uttar Pradesh (19.0%), West Bengal (18.9%) and Tamil Nadu (18.6%). It is below 10% in Kerala, Assam, Jharkhand and Gujarat. The largest gains since 2019-20 are in Bihar (+2.9 points), Haryana (+2.8) and Punjab (+2.6).

ST share is above the 7.5% reference point in Jharkhand, Chhattisgarh, Assam, Odisha, Rajasthan, Gujarat, Madhya Pradesh and Telangana. It is below 5% in 10 of the 20 larger states, including Uttar Pradesh (1.0%), Tamil Nadu (0.9%), Bihar (1.8%), West Bengal (3.8%) and Maharashtra (4.8%). State averages hide local gaps, and judging them needs Census ST shares, which are not in AISHE.

OBC share rose by 11.8 points in Kerala, 8.9 in Bihar and 6.8 in Tamil Nadu. Changes this size can come from how institutions report category, so we show them but do not interpret them.

How far the pipeline reaches: Ph.D. students per 100 PG students

A second state measure is the number of Ph.D. students for every 100 PG students. For India it is 5.9. It is 2.3 in Madhya Pradesh, 3.0 in Chhattisgarh, 3.1 in Bihar, 3.8 in Telangana and 4.0 in Uttar Pradesh, and it is about 10 in Assam (10.4), Punjab (10.4), Tamil Nadu (10.1) and Delhi (9.9).

This matters for inclusion because the states where the pipeline is narrowest for everyone are also big UG states. Uttar Pradesh has 19.0% SC students overall and only 4.0 Ph.D. students per 100 PG students. The ratio compares stocks and uses the state of the institution, not where a student comes from; students move to other states for Ph.D. programmes.

Women's share of Ph.D. students also rose in nearly every large state between 2019-20 and 2023-24, most in Jharkhand (+11.3 points), Uttarakhand (+9.5) and Tamil Nadu (+7.9), least in Madhya Pradesh (+0.3) and Delhi (+1.1). These come from the estimated state tables (Table 43), not from Table 11, and are not mixed with it.

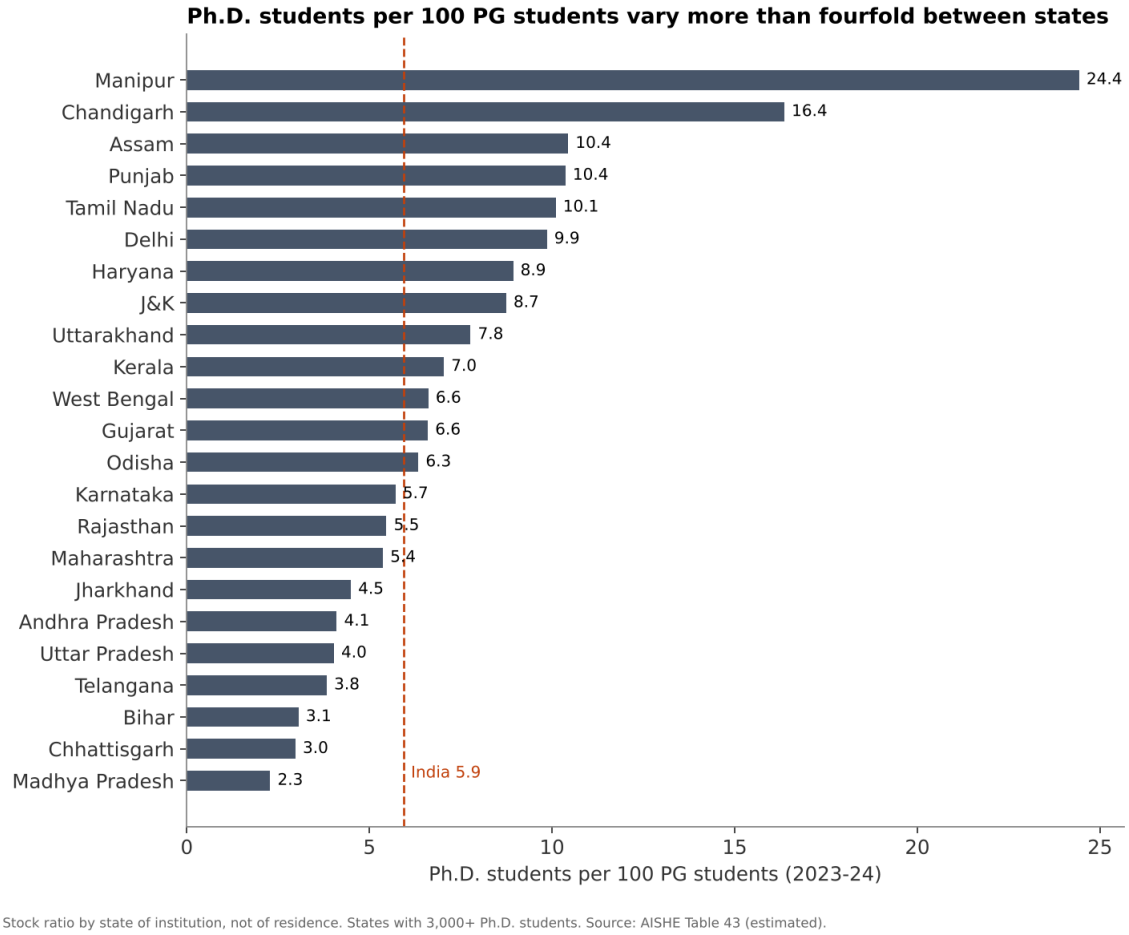


Figure 10: *Ph.D. students per 100 PG students, by state, 2023-24. States with at least 3,000 Ph.D. students. Dashed line: India, 5.9. Source: AISHE Table 43 (estimated).*

Do enrolled students finish? Out-turn by category

AISHE also reports out-turn, the number of students who pass out in a year, by category (Table 33a). It covers all levels together, so it cannot show Ph.D. completion. It does show whether SC, ST and OBC students leave the system with degrees at the rate they enter.

In 2023-24 the SC share of out-turn was 14.3% against 15.5% of enrolment, a ratio of 0.92. For ST it was 6.0% against 6.4% (0.94) and for OBC 38.9% against 40.1% (0.97). In 2021-22 the SC ratio was 0.85. It rose to 0.91 in 2022-23 and 0.92 in 2023-24.

Out-turn lags enrolment by the length of the course, and SC enrolment share has been rising, so part of the gap may be this lag and not a higher dropout. The data cannot separate the two. A cohort study (tracking admission batches) would be needed.

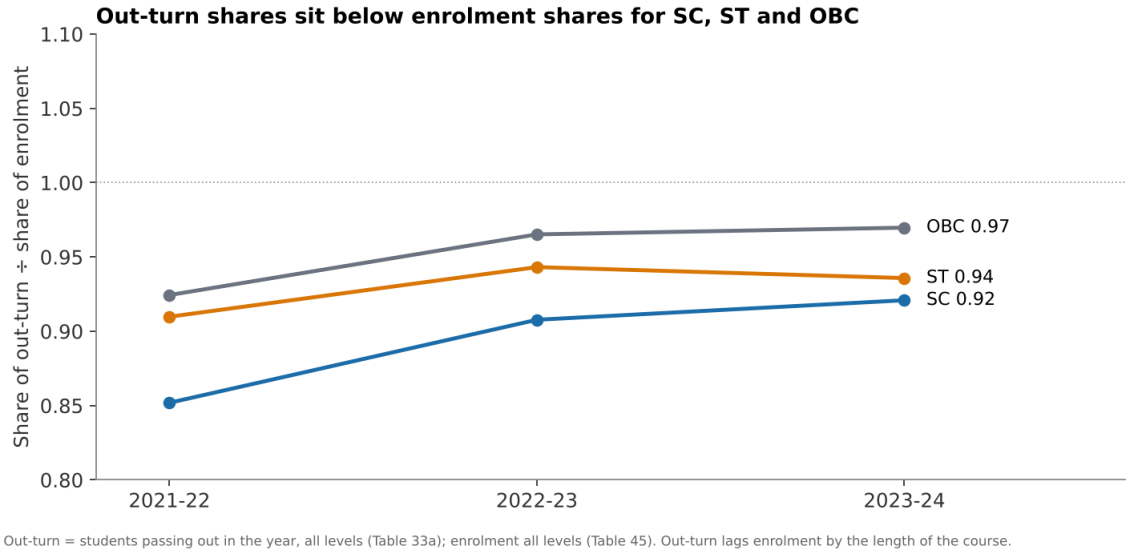


Figure 11: *Share of out-turn divided by share of enrolment, 2021-22 to 2023-24. Below 1.00 means the group is a smaller share of those passing out than of those enrolled. Source: AISHE Tables 33a and 45.*

Institutes of National Importance: SC and ST near the average, women far below it

Institutes of National Importance (INIs) had 13.4% SC and 6.5% ST students in 2023-24, close to the all-university average of 12.0% and 5.6%, but only 30.2% women against 44.8%. State private and private deemed universities have the lowest SC shares (7.4% and 5.6%), and private deemed universities the lowest ST share (1.3%).

INIs have improved since 2019-20: SC from 12.0% to 13.4%, ST from 5.5% to 6.5% and women from 24.7% to 30.2%. These figures cover teaching departments of universities and INIs, not affiliated colleges, and mix all levels. AISHE publishes no Ph.D.-level category split by university type.

Table 3: *Enrolment by type of university, all levels, 2023-24*

University type	Students	Women	SC	ST	OBC
Institutes of National Importance	417,087	30.2%	13.4%	6.5%	26.4%
Central universities	795,467	48.3%	13.2%	4.4%	18.0%
State public universities	3,183,668	50.2%	14.8%	7.2%	40.6%
State private universities	2,319,465	36.3%	7.4%	4.4%	24.6%
Deemed universities (private)	1,089,939	42.2%	5.6%	1.3%	34.3%
All university types	10,858,898	44.8%	12.0%	5.6%	32.2%

Source: AISHE 2023-24, Table 27(a), actual responses.

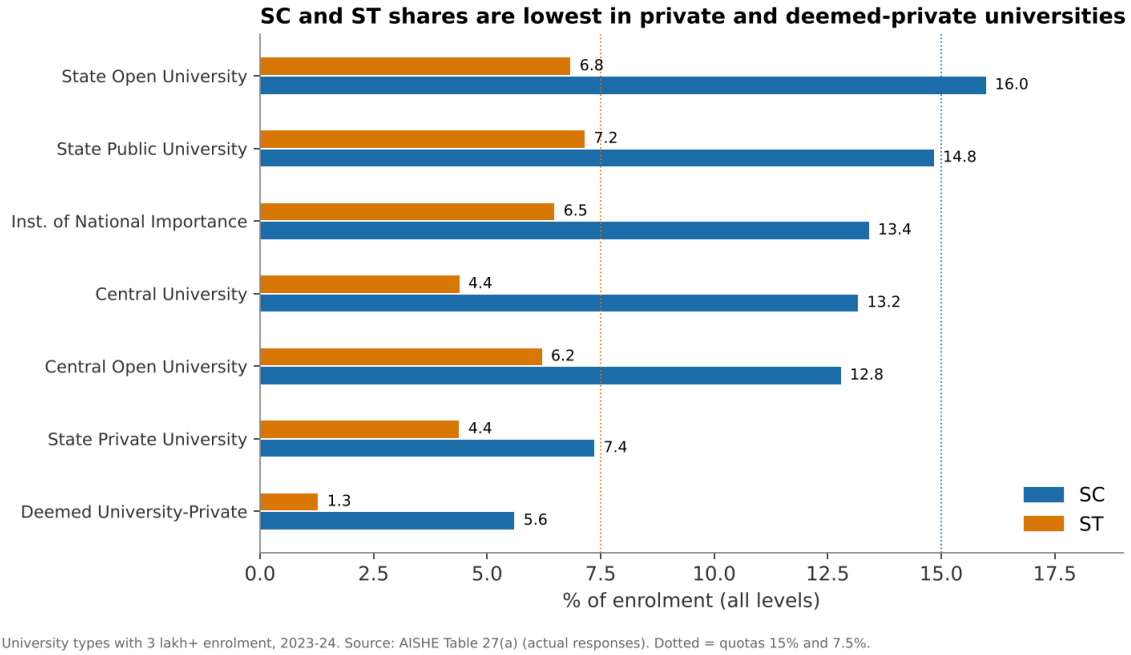


Figure 12: *SC and ST share by type of university, 2023-24. University types with at least 3 lakh students. Dotted lines mark 15% and 7.5%. Source: AISHE Table 27(a), actual responses.*

State open and state public universities have the highest SC shares (16.0% and 14.8%). Together, state private and private deemed universities enrol 31% of the students in Table 27(a) but have 7.4% and 5.6% SC students. Central universities, where reservation in admission is mandated, are below the 15%, 7.5% and 27% reference points on all three (SC 13.2%, ST 4.4%, OBC 18.0%) when all levels are counted. These are shares of enrolment, not of admissions to reserved seats, so they are a signal to look closer.

For women, INIs are a different case from the states. Women are 26.5% of INI UG students in 2023-24 but 34.6% of INI Ph.D. students. Both are low against all universities (41.3% at UG, 48.6% at Ph.D.). Women's Ph.D. share in INIs rose 2.4 points in five years (32.2% to 34.6%) while in all universities it rose 4.6 (44.0% to 48.6%), so the gap grew from 11.8 to 14.0 points.

AISHE reports INIs together, so this cannot say whether the gap is largest in IITs, IISERs or others.

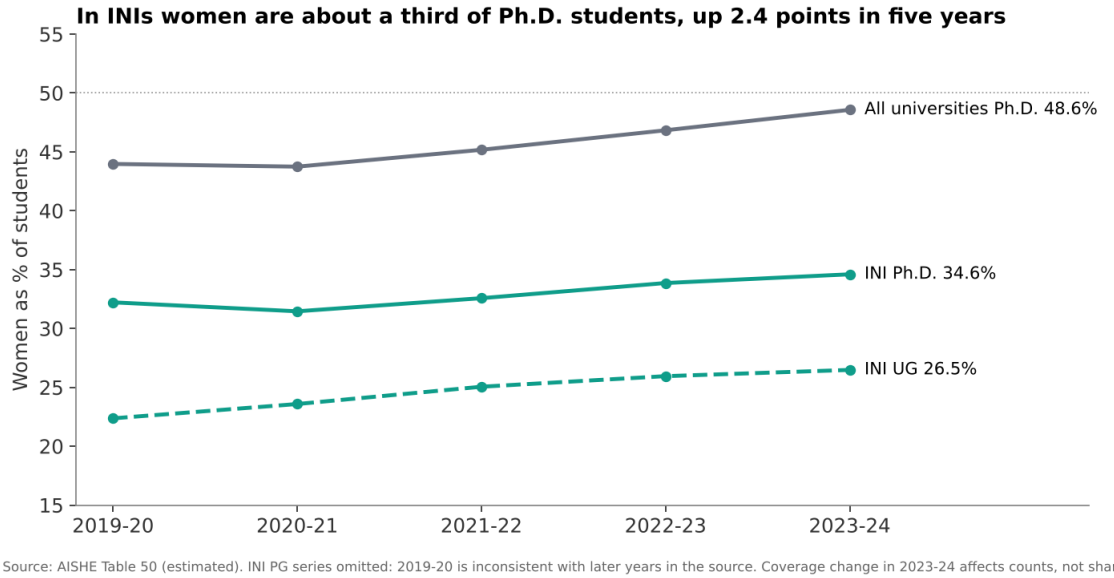


Figure 13: *Women in Institutes of National Importance and all universities, 2019-20 to 2023-24. INI PG omitted because the 2019-20 split is inconsistent in the source. Source: AISHE Table 50.*

What the data cannot show

- **SC/ST Ph.D. students by state:** not published by AISHE. State SC/ST figures are all levels combined; state Ph.D. figures are by gender only.
- **Individual institutes:** AISHE groups IITs, NITs, IISERs and others as one category. Institute-wise data come from NIRF submissions, Parliament replies and RTI.
- **SC/ST women in the Ph.D. pipeline by institute:** not public in any source we found. An RTI request per institute is the realistic route and takes up to 30 days.
- **Counts versus shares:** Ph.D. counts jump 47% in 2023-24, probably from wider survey coverage. We compare shares across years.
- **Actual and estimated tables:** national category figures (Table 11) are actual responses; state and university-type gender figures are AISHE estimates. They differ slightly and are not mixed in one ratio.
- **Small states:** states and UTs with fewer than 1,000 Ph.D. students (Ladakh, Lakshadweep, Andaman and Nicobar, Dadra and Nagar Haveli and Daman and Diu, Tripura) give unstable percentages and hence are excluded from the study
- **Out-turn:** Table 33a covers all levels together and lags enrolment, so it is not a Ph.D. completion rate. The "missing students" figures are an illustration of scale, not a target.
- **Reference points, not verdicts:** Ph.D. enrolment includes part-time and sponsored scholars and institutions outside reservation rules. A gap below 15% or 7.5% may have several reasons including low number of applications in the first place, this by itself, is not proof of non-compliance.

Recommendations

These are proposals from our reading of the data, written as a list for the people who publish and use higher-education data and for other teams.

- **The SC, ST and OBC enrollment data can be published by level and by state.** The published tables give category by programme only nationally, and category by state only for all levels combined.
- **The Ph.D. admissions and enrolment can be published by category and gender for each institute every year,** including IISERs and IITs, in a downloadable table.
- **The focus should be to support the step from PG to Ph.D.,** where the SC share falls from 13.7% to 10.0% and the ST share from 6.2% to 4.1%.
- **Attention needs to be paid to SC women and to engineering.** SC women are 44.4% of SC Ph.D. students while women are 50.1% of all Ph.D. students, and women are 37% of Ph.D. students in engineering.
- **The states where women's Ph.D. share falls well below their UG share need to be looked at** (West Bengal, Jharkhand, Telangana, Bihar, Odisha) for the local reasons behind the drop.
- **The out-turn by level and category can be published,** so that completion at the Ph.D. level can be measured.

Literature and sources

- Department of Higher Education, Ministry of Education, Government of India. *All India Survey on Higher Education, Final Reports 2019-20, 2021-22, 2022-23, 2023-24* (Excel). aishe.gov.in/aishe-final-report
- AISHE tables used: Table 6 and 43 (level by gender, by state), Table 11 (programme-wise enrolment by category), Table 13 (Ph.D. by discipline), Table 27 (university type by category), Table 45 (category by state, five years), Table 33a (out-turn by category), Table 50 (university type by level and gender).
- iGEM Paris-Bettencourt 2013. *Gender Study* (Human Practice). 2013.igem.org. An earlier iGEM data study of gender balance, whose structure this page follows.

Attributions

This study was designed and carried out by Dixita Adhikari of the IISER Kolkata iGEM team.

All data were taken from public AISHE reports. Figures were drawn from the AISHE Excel files; all numbers on this page are in the downloadable database.